AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 4 · STUDENT LESSON

Python and Table Readiness

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

AI-3 Week 23: initialise, loop and indent Python correctly.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Specify a table schema. Try the recall before opening notes.

Your learning target

I can specify a table **schema**.

I can check **type**s and **unit**s as well as names.

I can distinguish structural validity from truth.

Remember before reading

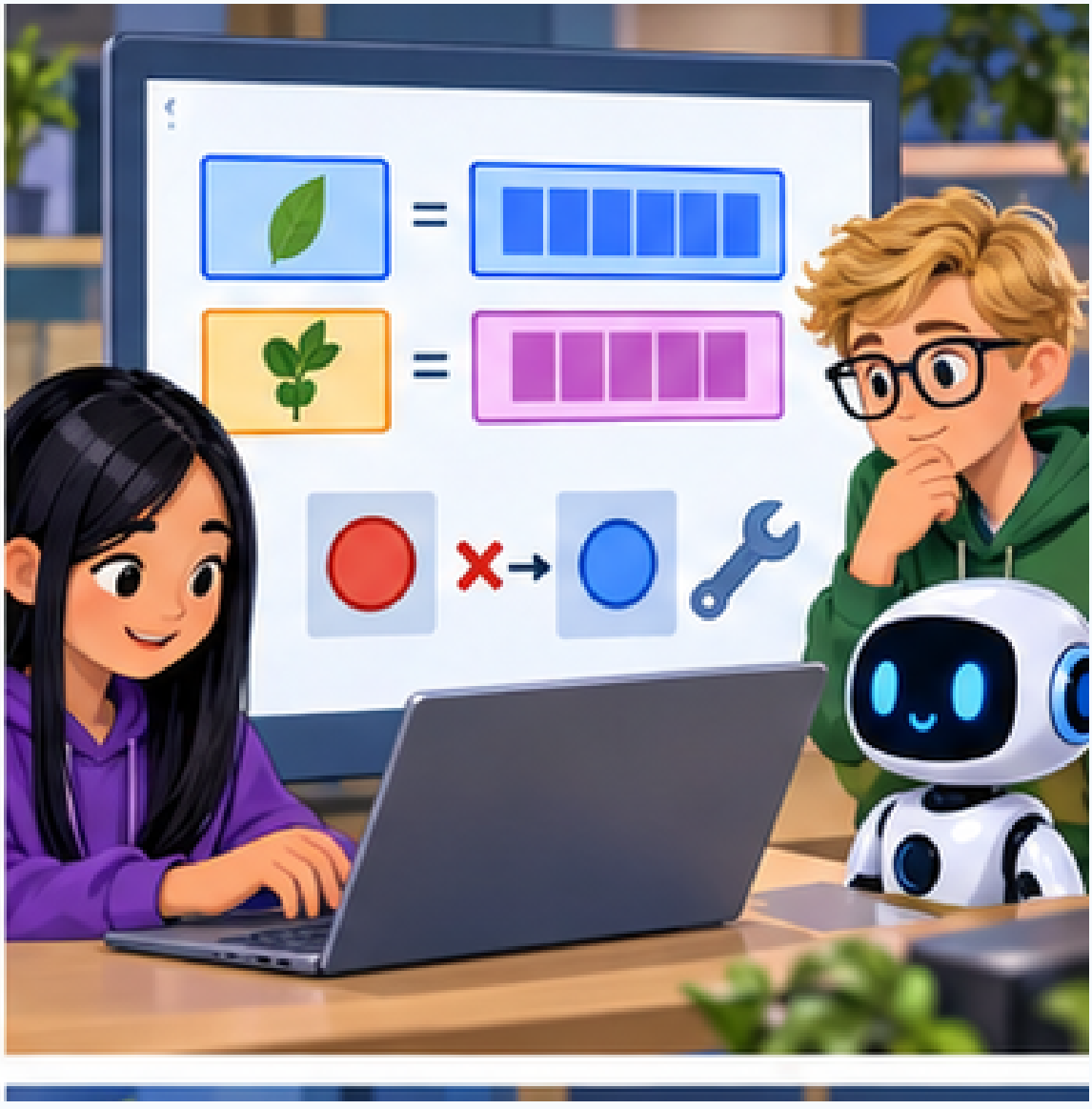
Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Classification or Regression?. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from The ML Lifecycle. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Python **variable**s, **type**s, **expression**s, rows, and columns give a reproducible representation of data and transformations.



Your target

LEARNING TARGET Trace mean for [4,8,12] using sum(values)/len(values). What should the program do for []? Explain why returning 0 silently is misleading.

Builder name and experiment date:

Readiness for the connected explanation

What does row["used"] mean?

Readiness for File input and output

Is "12" immediately usable as an integer count?

My readiness explanation

Readiness for Exceptions and recovery

Which error can int("twelve") raise?

My readiness explanation

This week’s end goal

By the end, I can explain python and table readiness using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

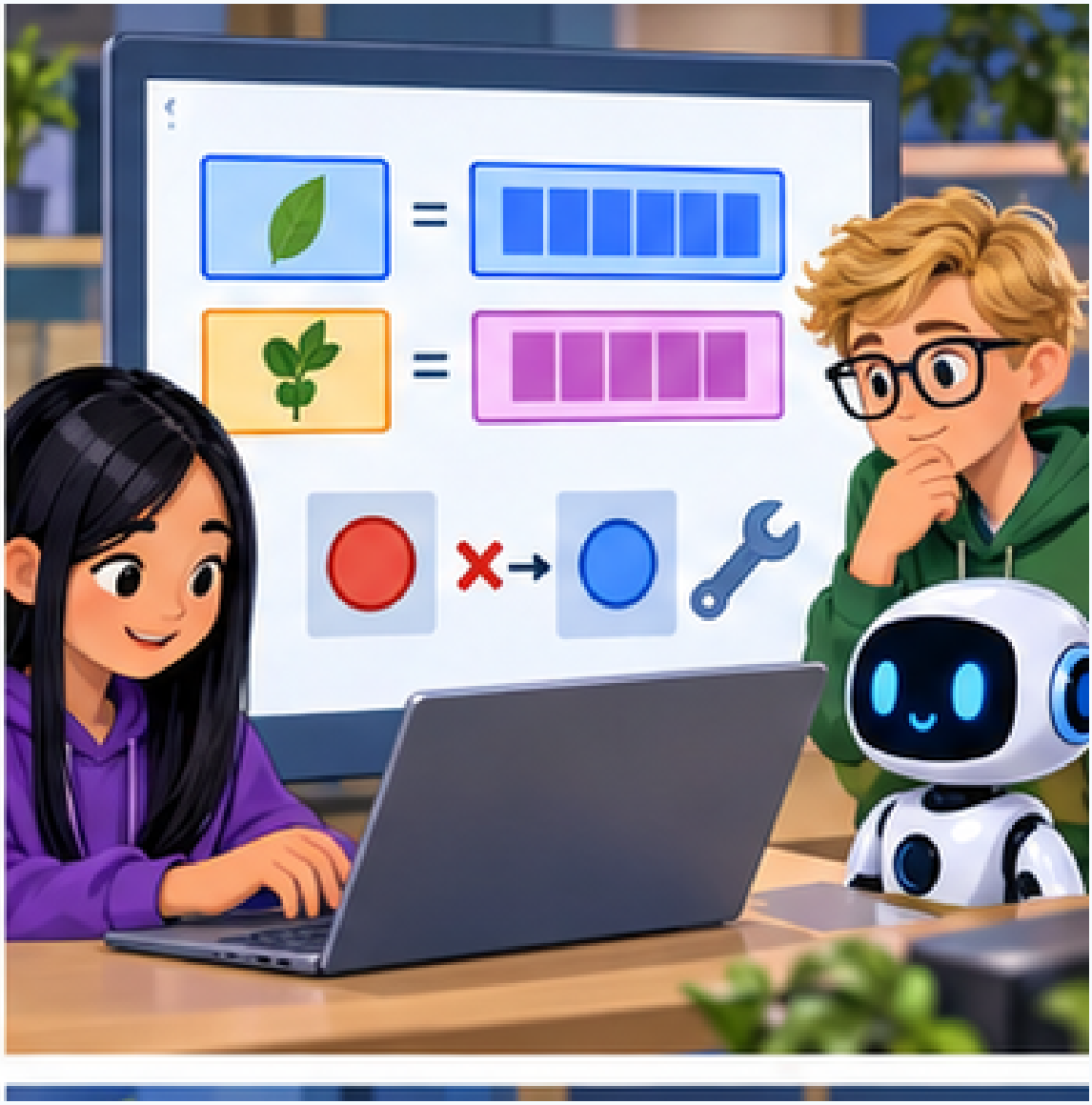
Notice the evidence

> Investigate the supplied Python and Table Readiness case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use python and table readiness to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

A fictional classroom logs the duration of each activity. Before comparing durations, define a **schema**: activity is nonempty text, minutes is an integer from 0 through 60, and each row describes one activity session.

These decisions make data interpretable.

Apply the connected idea: File input and output

Use a teacher-provided local sandbox folder, never a real personal **file**. Create durations.txt with two lines: 12 and 8. The goal is to read the text, convert it to integers and report the total.

Keep the original **file** unchanged for this activity.

Apply the connected idea: Exceptions and recovery

A small parser receives duration text. Its contract permits a string representing a nonnegative integer up to 60. Invalid numeric text should return a clear **status** rather than crash or pretend that the duration was zero.

Test both the successful and failing **path**s.

STUDENT LESSON · 3 OF 18

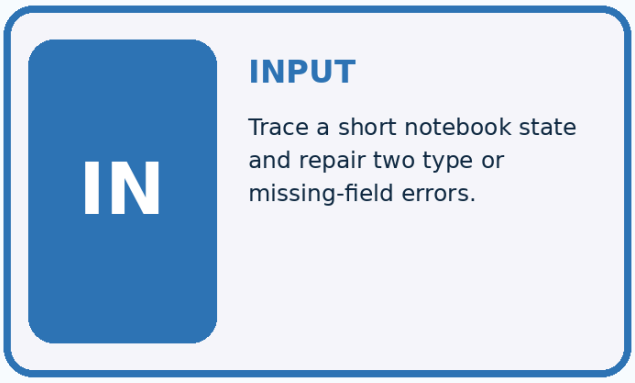
See how it works

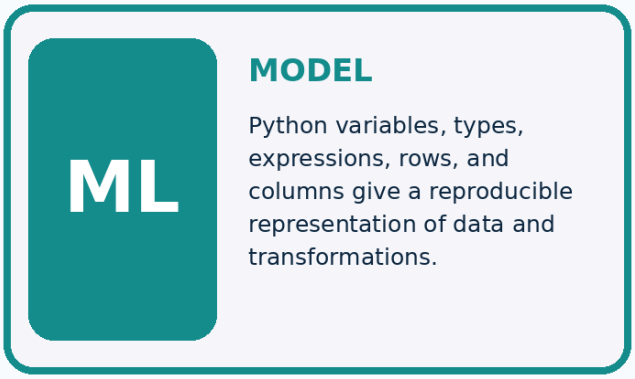
> Follow the mechanism for Python and Table Readiness. Connect each labelled input, deciding step and result.

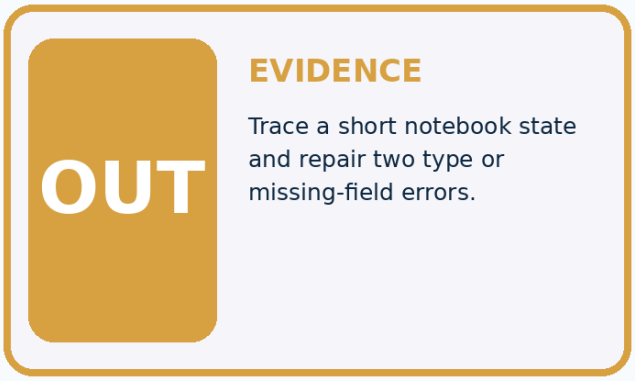
ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Trace a short notebook state and repair two **type** or missing-field errors.

MODEL IDEA Python **variable**s, **type**s, **expression**s, rows, and columns give a reproducible representation of data and transformations.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD widths = [4.2, 5.0, 6.1] stores three decimal measurements. mean_width = sum(widths) / len(widths) evaluates to 5.1.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

```
rows = [{"activity": "Reading", "minutes": 12},
        {"activity": "Sorting", "minutes": 0}]
for row in rows:
    valid = (type(row["activity"]) is str
             and len(row["activity"]) > 0
             and type(row["minutes"]) is int
             and 0 <= row["minutes"] <= 60)
    print(valid)
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
activity	Nonempty text	Names the session activity
minutes	Integer 0 through 60	Duration in minutes

Read each row from left to right and connect the evidence to the stated rule.

The worked script prints True twice. **type**(value) is int checks the exact **type** here and avoids accepting a Boolean as a count. The chained comparison checks both boundaries.

The example assumes both keys exist; next week handles missing fields. A zero duration is allowed by this written **schema**.

Check a row in separate steps

A **schema** is a checklist for what a row must contain. First ask whether the activity name is text and is not empty. Next ask whether minutes is an integer.

Then ask whether that integer lies between 0 and 60, including both ends. Keep these checks separate when explaining a rejection.

The number 12 and the text "12" look similar on a page, but only the number meets this lesson’s minutes rule.

Apply the connected idea: File input and output

```
with open("durations.txt", "r", encoding="utf-8") as handle:
    lines = handle.read().splitlines()
values = []
for line in lines:
    values.append(int(line))
print(sum(values))
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Local file	12 then 8 on separate lines	Read as text
Conversion	"12" to 12 and "8" to 8	Integer values
Summary	12 + 8	Print 20

Read each row from left to right and connect the evidence to the stated rule.

The mode r means read. with closes the **file** when the block finishes, including when an error occurs. splitlines separates lines without retaining newline characters.

This example assumes each line is a valid integer text; an empty line or twelve raises ValueError during int conversion.

Separate file text from numbers

The **file** contains characters, not ready-made Python integers. Reading gives text; splitlines separates the lines; int converts each valid line into a number. The lines "12" and "8" become 12 and 8, so their total is 20.

Keep those three stages visible in your trace. A line such as "twelve" fails conversion, and a missing **file** is a different problem. Neither failure means the observed duration was zero.

Apply the connected idea: Exceptions and recovery

```
def parse_minutes(text):
    try:
        value = int(text)
    except ValueError:
        return {"status": "invalid text", "value": None}
    if value < 0 or value > 60:
```

```
        return {"status": "out of range", "value": None}
    return {"status": "ok", "value": value}

for text in ["12", "zero", "0", "61"]:
    print(parse_minutes(text))
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
"12" or "0"	Conversion and range pass	ok with numeric value
"zero"	Conversion fails	invalid text with None
"61"	Integer outside range	out of range with None

Read each row from left to right and connect the evidence to the stated rule.

try covers the operation expected to fail; except ValueError handles that known conversion failure. Returning early ends the failing **path**. This contract expects a string: it does not claim to handle every possible object.

Spaces around an integer are accepted by int, and that behavior should be documented.

Give each failure an honest result

Trace the function as a decision **path**. First try to turn the text into an integer. If that conversion raises ValueError, return invalid text and None. If conversion succeeds, check the range.

A value outside 0 through 60 returns out of range and None. Only a value passing both stages returns ok and its number. None means there is no accepted number; it must not be added to a duration total.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

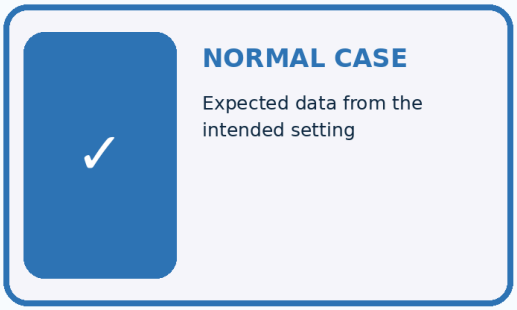
Compare the cases

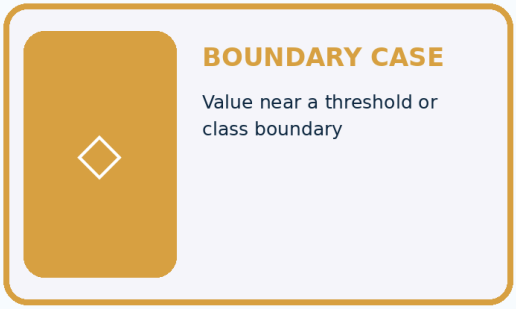
> Compare the supplied cases for Python and Table Readiness. Identify the detail that changes the conclusion.

EVALUATION DESIGN

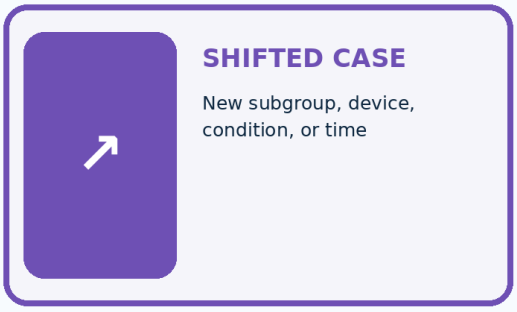
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: S changes from 0 to None.

Predicted result

Prediction: Mean becomes (2+6)/2=4 cm with 2/4 measured.

Why the result follows

Reason: Removing a measured zero changes both the denominator and the coverage.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Python and Table Readiness to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.


1
VARIABLE
name bound to a value
2
DATA TYPE
kind of value and valid operations
3
DATAFRAME
table represented in code
4
EXPRESSION
code that computes a value

Term	Precise meaning	My ML example
variable	name bound to a value	
data type	kind of value and valid operations	
dataframe	table represented in code	
expression	code that computes a value	

> MISCONCEPTION TO REPAIR If notebook cells eventually run, execution order and **data type**s do not matter.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Schema	Rules for the fields and values in records
Type	A category such as integer or text
Unit	The quantity scale attached to a number
Constraint	A rule a valid record must satisfy

Terms for File input and output

Term	Meaning
File	Stored data with a name and location
Path	A location used to find a file
Encoding	A rule for representing text as bytes
Context manager	A with block that manages a resource such as closing a file

Terms for Exceptions and recovery

Term	Meaning
Exception	A signal that an operation could not complete normally
Handler	Code that responds to a specified exception
Recovery	A defined next step after a failure
Status	A label stating the outcome of a processing step

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Python and Table Readiness, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 4 • WORKED EVIDENCE

1. widths = [4.2, 5.0, 6.1] stores three decimal measurements.
2. mean_width = sum(widths) / len(widths) evaluates to 5.1.
3. Adding 'unknown' to the numeric list causes a type problem.
4. Clean or explicitly encode values, rerun top to bottom, and record the result.

Step 1

STEP 1 widths = [4.2, 5.0, 6.1] stores three decimal measurements.

Step 2

STEP 2 mean_width = sum(widths) / len(widths) evaluates to 5.1.

Step 3

STEP 3 Adding 'unknown' to the numeric list causes a **type** problem.

Step 4

STEP 4 Clean or explicitly encode values, rerun top to bottom, and record the result.

CHECK Verify the denominator, **unit**, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Would minutes of 61, "12" and 12 each pass? Explain the type distinction.

Apply the connected idea: File input and output

If the file contains 0, 3 and 7 on separate lines, predict the values and total.

Apply the connected idea: Exceptions and recovery

Predict statuses and values for "-1", " 5 " and "3.5".

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Python and Table Readiness

Inspect a small table before asking Python to compute an average.

The facts we will use

> **Supplied facts**
>
> Plant heights in cm: P=2, Q=6, R=None, S=0. None means unmeasured; zero is a recorded valid height.

> **Supplied facts**
>
> All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/4

Python and Table Readiness

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Check records

Four rows contain three measured values and one missing value.

2. Calculate mean

Measured sum is 2+6+0=8; mean is 8/3 cm, approximately 2.67 cm.

3. Report coverage

Report 3/4 measured, retaining the missing record separately.

CHANGE ONE THING

S changes from 0 to None.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Check records	Four rows contain three measured values and one missing value.
2. Calculate mean	Measured sum is 2+6+0=8; mean is 8/3 cm, approximately 2.67 cm.
3. Report coverage	Report 3/4 measured, retaining the missing record separately.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Report 3/4 measured, retaining the missing record separately. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

The measured-row mean may not represent the missing plants; investigate why those

readings are absent.

Keep the stated limit

Read the labelled picture: Why should the missing value not silently become zero? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: S changes from 0 to None. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Python and Table Readiness claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: If notebook cells eventually run, execution order and data types do not matter. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE If notebook cells eventually run, execution order and **data type**s do not matter.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

A row can satisfy every **schema** rule and still be a wrong **observation**. **Schema** validation checks our stated structure and **constraint**s, not whether someone measured honestly or selected a representative sample.

A row says Reading, 12 and passes. Does that establish the session really lasted 12 minutes?

Apply the connected idea: File input and output

File reading does not establish that the values are valid durations. After conversion, apply the **schema** and record any rejected lines. Do not upload personal records to an external tool for this exercise.

The file contains twelve on one line. Why should the program not silently turn it into zero?

Apply the connected idea: Exceptions and recovery

Recovery must fit the task. A caller may ask for corrected **input** or exclude the row with a recorded reason. It must not use None as a number or claim that rejected records were successfully **process**ed.

A batch has 8 valid and 2 invalid lines. What should its report include?

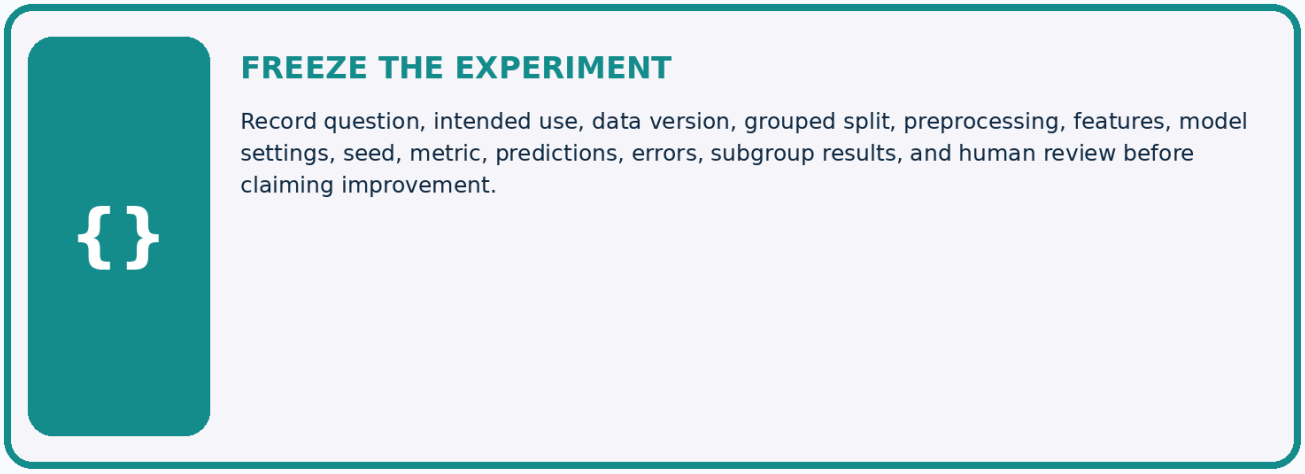
Practise together

> Practise Python and Table Readiness with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

Question to explain

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + **unit**s + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

If one row records 120 seconds in the minutes field, a simple numeric total becomes misleading. Correct the source or explicitly convert and document it; renaming a column alone does not convert values.

Apply the connected idea: File input and output

If Python cannot find durations.txt, it raises **File**NotFoundError. Check the current working folder and exact **file**name.

Do not switch to write mode to solve a missing **input**: opening an existing **file** with w would replace its contents.

Apply the connected idea: Exceptions and recovery

A bare except that returns 0 hides unrelated programming errors and corrupts missing values. Catch the expected **exception**, preserve an error **status** and keep the unknown value as None.

If **file** reading is added, handle **File**NotFoundError separately with an actionable message.

Practise with less help: Classify records

Use workbook W2 for the connected example “Rows columns and **schema**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Rows columns and **schema**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: File input and output

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

Practise the connected topic: Exceptions and recovery

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

Explain the picture

Why should the missing value not silently become zero? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

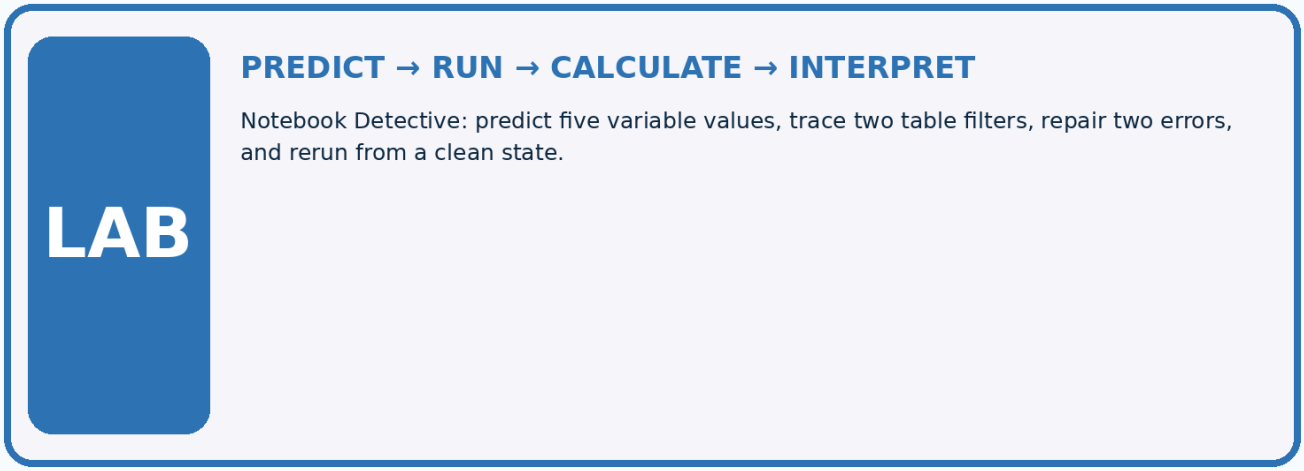
Run the investigation

> Investigate Python and Table Readiness. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Notebook Detective: predict five **variable** values, trace two table filters, repair two errors, and rerun from a clean state.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

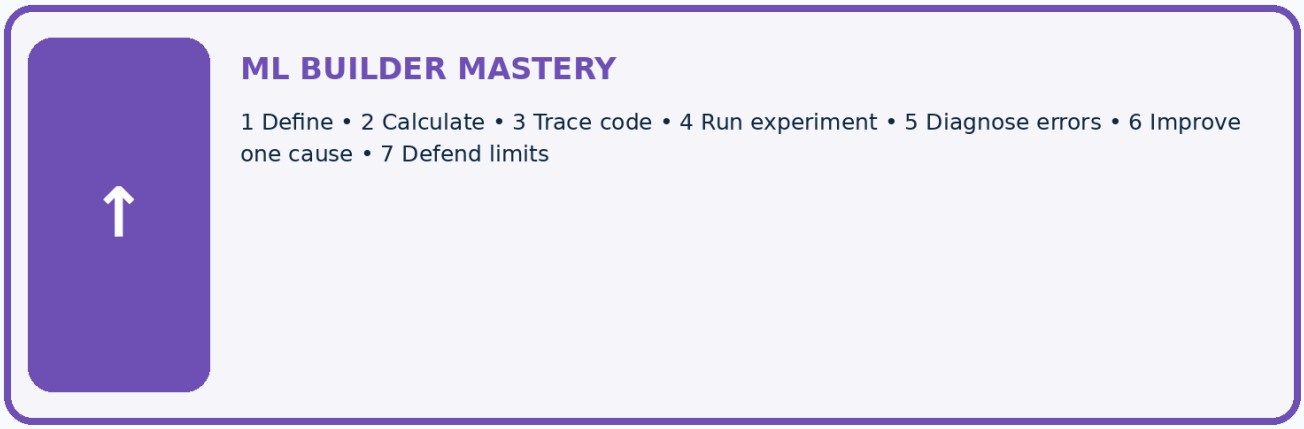
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Python and Table Readiness. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **variable**, **data type**, **dataframe**, **expression**.

CALCULATE Reproduce the worked **evidence** with **unit**s and denominator.

TRACE Explain the data, code or decision **path**.

DIAGNOSE If notebook cells eventually run, execution order and **data type**s do not matter.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.

RESPONSIBLE PLANT CLASSIFIER

Create a tiny five-row plant table with explicit column names, types, units, and a reproducible summary calculation.

THIS WEEK'S CHECKPOINT Create a tiny five-row plant table with explicit column names, **type**s, **unit**s, and a reproducible summary calculation.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom proto**type** for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Trace a short notebook state and repair two **type** or missing-field errors.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: Inspect Before Modeling.

Study a complete case

> Revisit the complete Python and Table Readiness case for supported practice. Repeated examples do not become fresh assessments.

Week 4 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Python lengths=[4,6,8]; total=sum(lengths); mean=total/len(lengths).

The question

Trace and handle an empty list.

Worked explanation

Total=18, count=3, mean=6.0. Guard the empty case against division by zero; no mean exists without **observation**s.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Trace the calculation and its empty case

Evidence or stage	Value or result
lengths	[4,6,8]
sum	18
count	3
mean	6.0
empty list	No mean, guard division by zero

Python lengths=[4,6,8]; total=sum(lengths); mean=total/len(lengths).

Trace and handle an empty list.

Total=18, count=3, mean=6.0. Guard the empty case against division by zero; no mean exists without **observation**s.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Changed independent case

Supplied values are3,7,14. Compute their mean. What should a function report for an empty list? How should it handle a missing fourth record?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 4 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Trace Python: lengths = [4, 7, 9]; selected = lengths[1]; total = selected + 3. Give selected and total.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Create a tiny five-row plant table with explicit column names, **type**s, **unit**s, and a reproducible summary calculation.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Trace mean for [4,8,12] using sum(values)/len(values). What should the program do for []? Explain why returning 0 silently is misleading.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

Your lesson route

> Use this map to navigate Python and Table Readiness, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Inspect a small table before asking Python to compute an average.

By the end, I can	How I will show it
Specify a table schema	Explain the deciding step, show a trace or test, and state one limit.
Check types and units as well as names	Explain the deciding step, show a trace or test, and state one limit.
Distinguish structural validity from truth	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

Finish and return

> Explain Python and Table Readiness without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Specify a table **schema**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Inspect a small table before asking Python to compute an average.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Report 3/4 measured, retaining the missing record separately.

1. Check records

2. Report coverage

3. Calculate mean

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report coverage Report 3/4 measured, retaining the missing record separately. Read the picture in order. Report 3/4 measured, retaining the missing record separately.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: S changes from 0 to None. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Mean becomes (2+6)/2=4 cm with 2/4 measured.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Mean becomes (2+6)/2=4 cm with 2/4 measured. Removing a measured zero changes both the denominator and the coverage.

Repair a missing building block

Review Classification or Regression? (ai-4/3). Goal: Identify a task from the meaning of its target. Short practice: Did regression MAE improve when the boundary changed?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review The ML Lifecycle (ai-4/1). Goal: Distinguish **fitting** a rule from applying a fixed rule. Short practice: Should the team silently retune on the changed test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

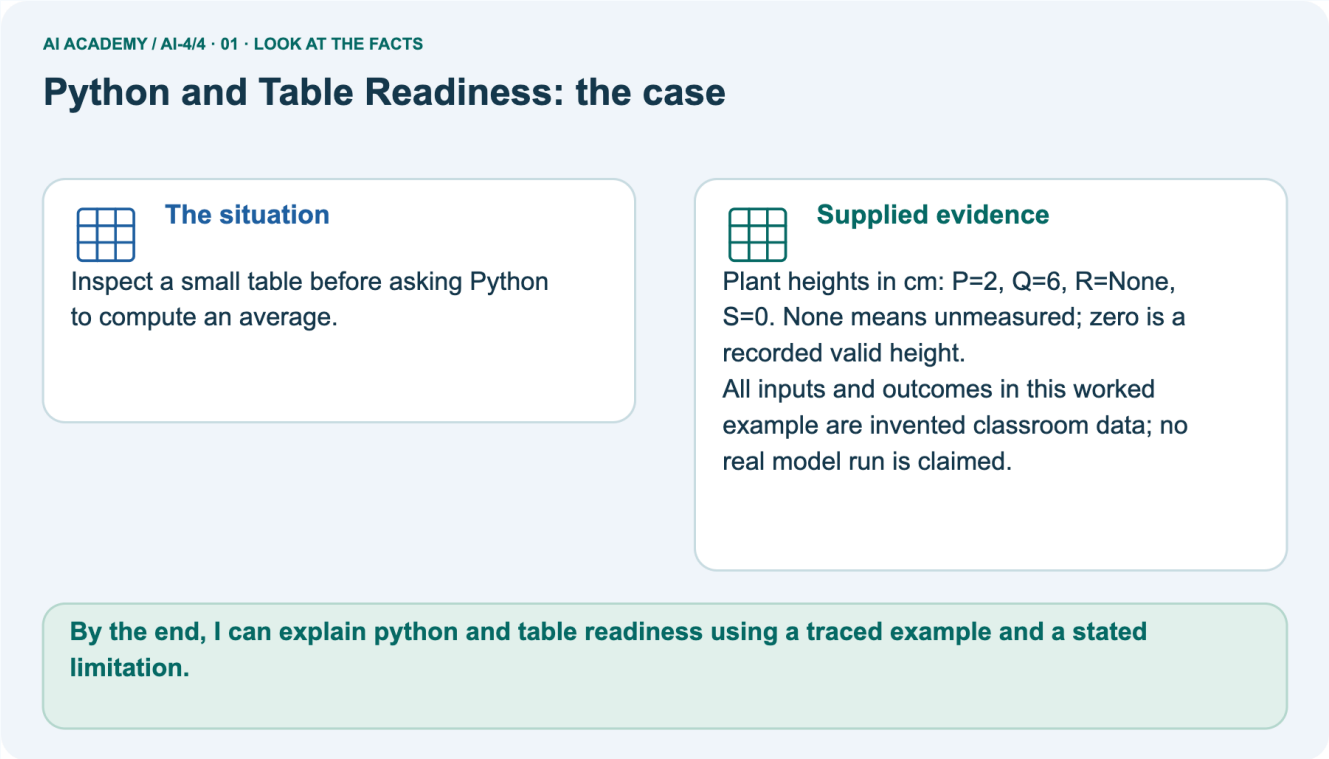
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Python and Table Readiness graphic using its labels, deciding evidence and relevant limitation.

01. Python and Table Readiness: the case



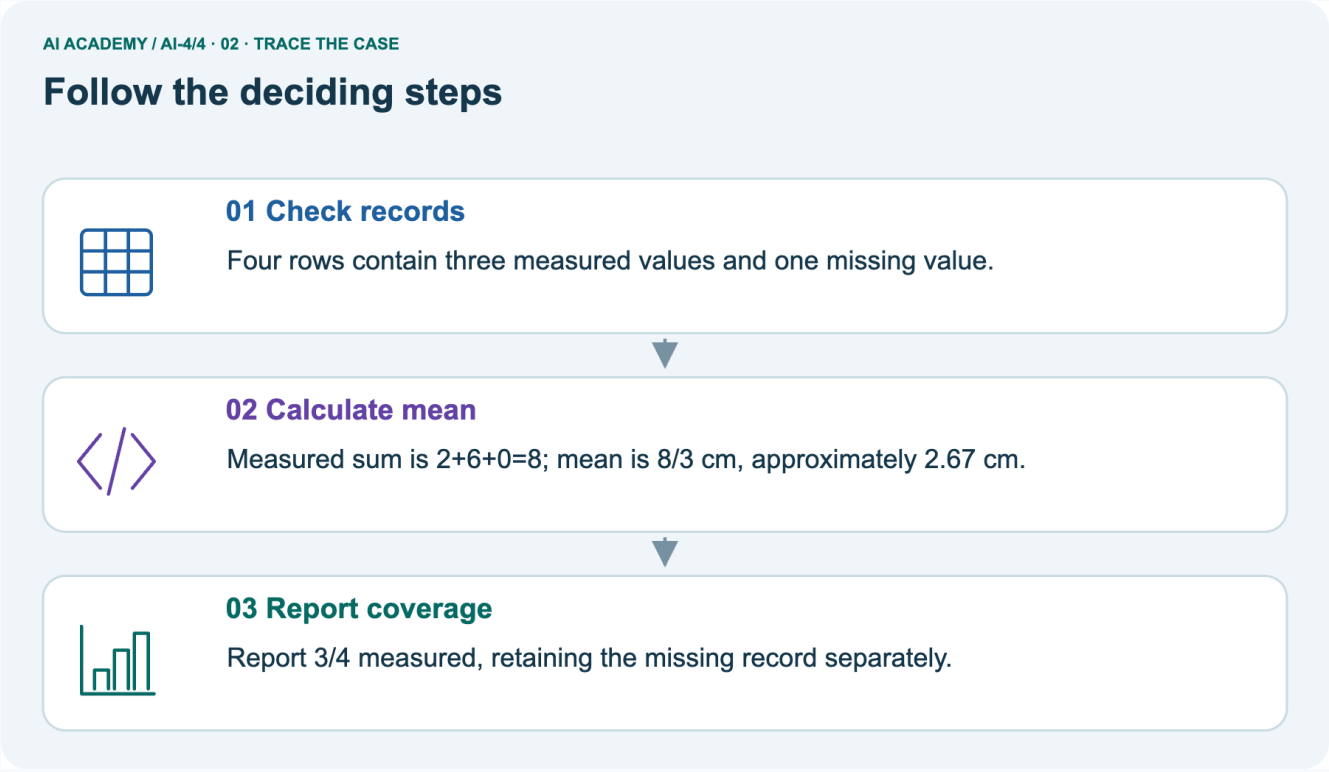
Follow the labelled picture with the matching explanation. Open the vector diagram

Inspect a small table before asking Python to compute an average. Supplied facts: Plant heights in cm: P=2, Q=6, R=None, S=0. None means unmeasured; zero is a recorded valid height.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

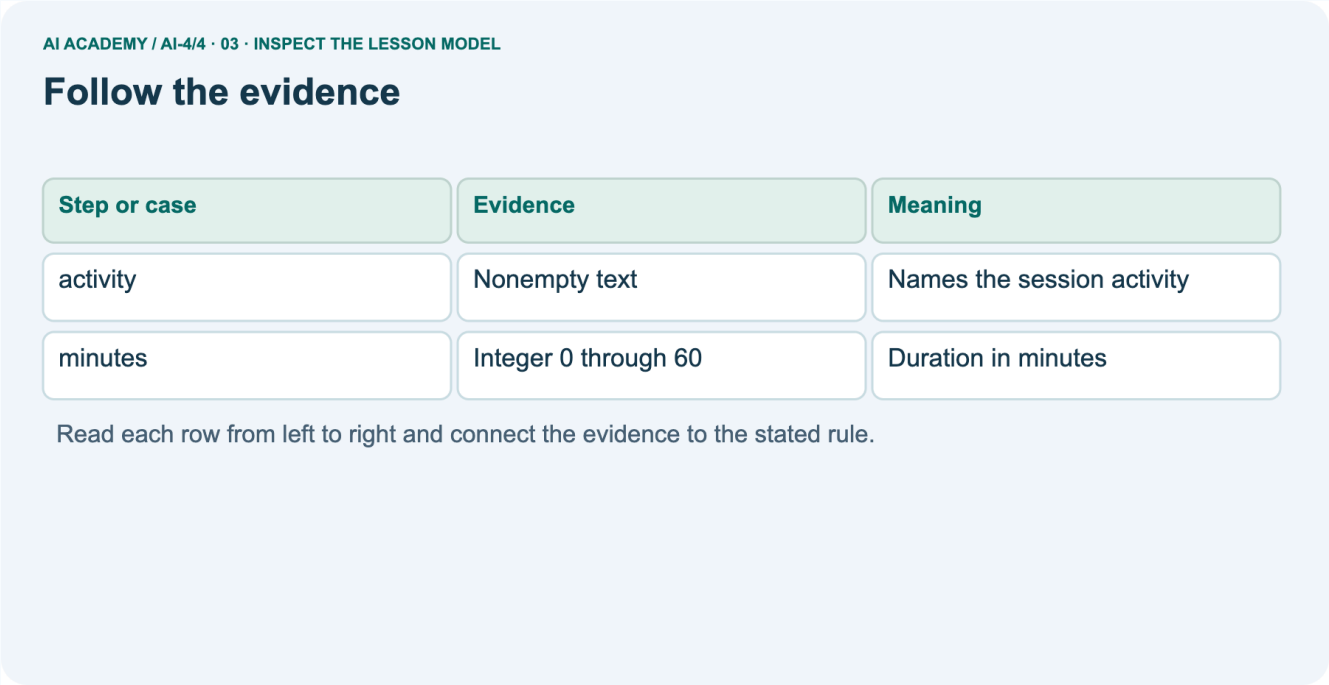


Follow the labelled picture with the matching explanation. Open the vector diagram

Check records: Four rows contain three measured values and one missing value. → Calculate mean: Measured sum is 2+6+0=8; mean is 8/3 cm, approximately 2.67 cm. → Report coverage: Report 3/4 measured, retaining the missing record separately.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Follow the evidence

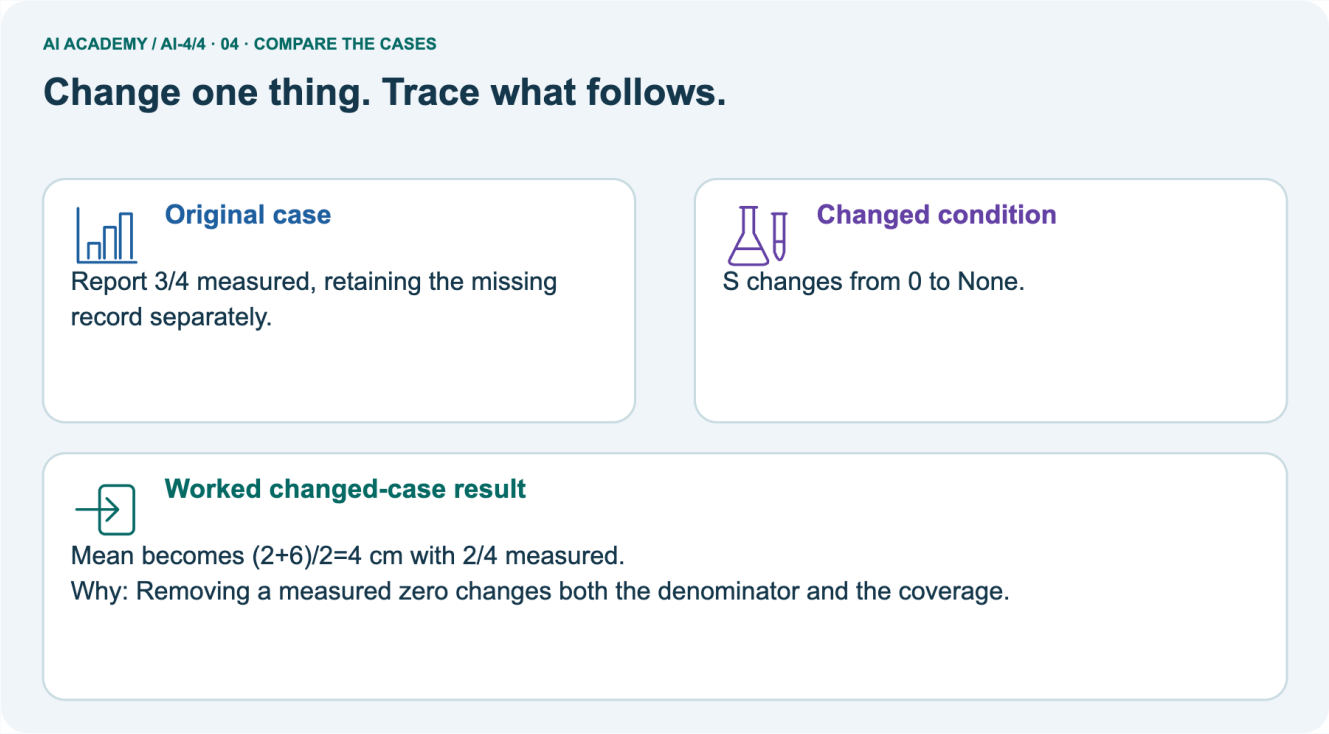


Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. activity | Nonempty text | Names the session activity. minutes | Integer 0 through 60 | Duration in minutes

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



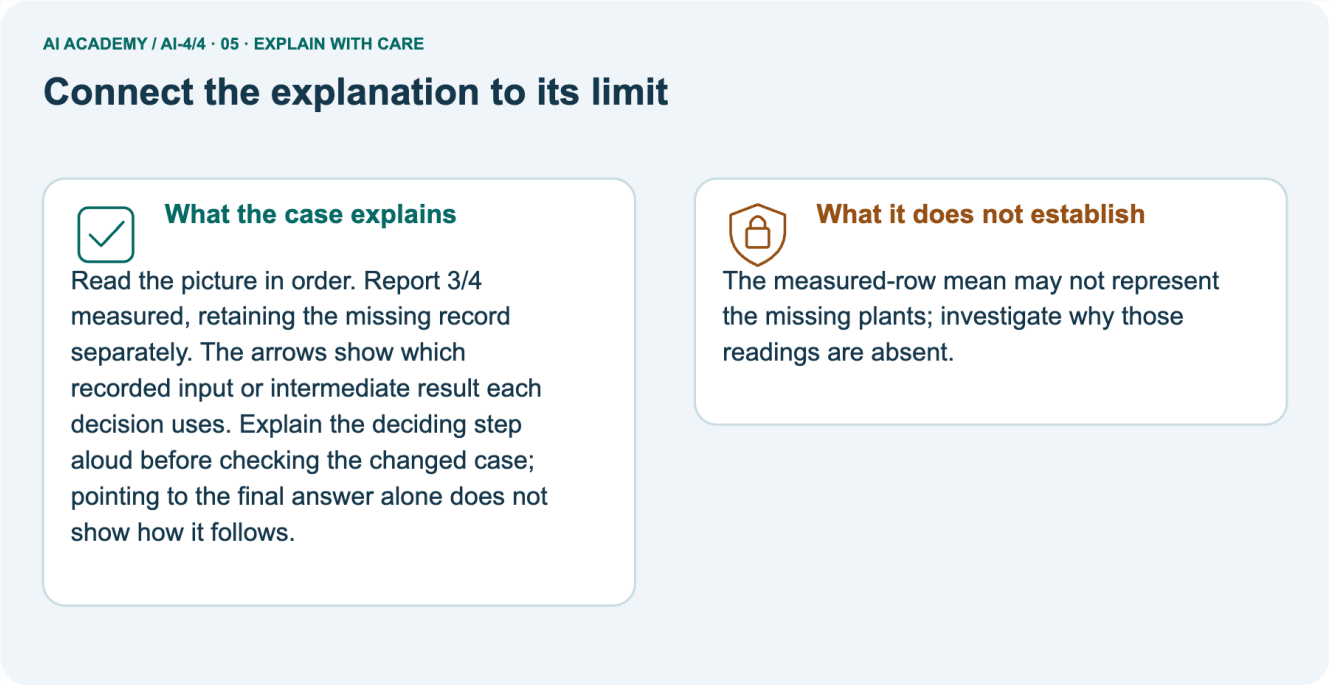
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Report 3/4 measured, retaining the missing record separately. Changed condition: S changes from 0 to None. Expected result: Mean becomes (2+6)/2=4 cm with 2/4 measured.

Reason: Removing a measured zero changes both the denominator and the coverage.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

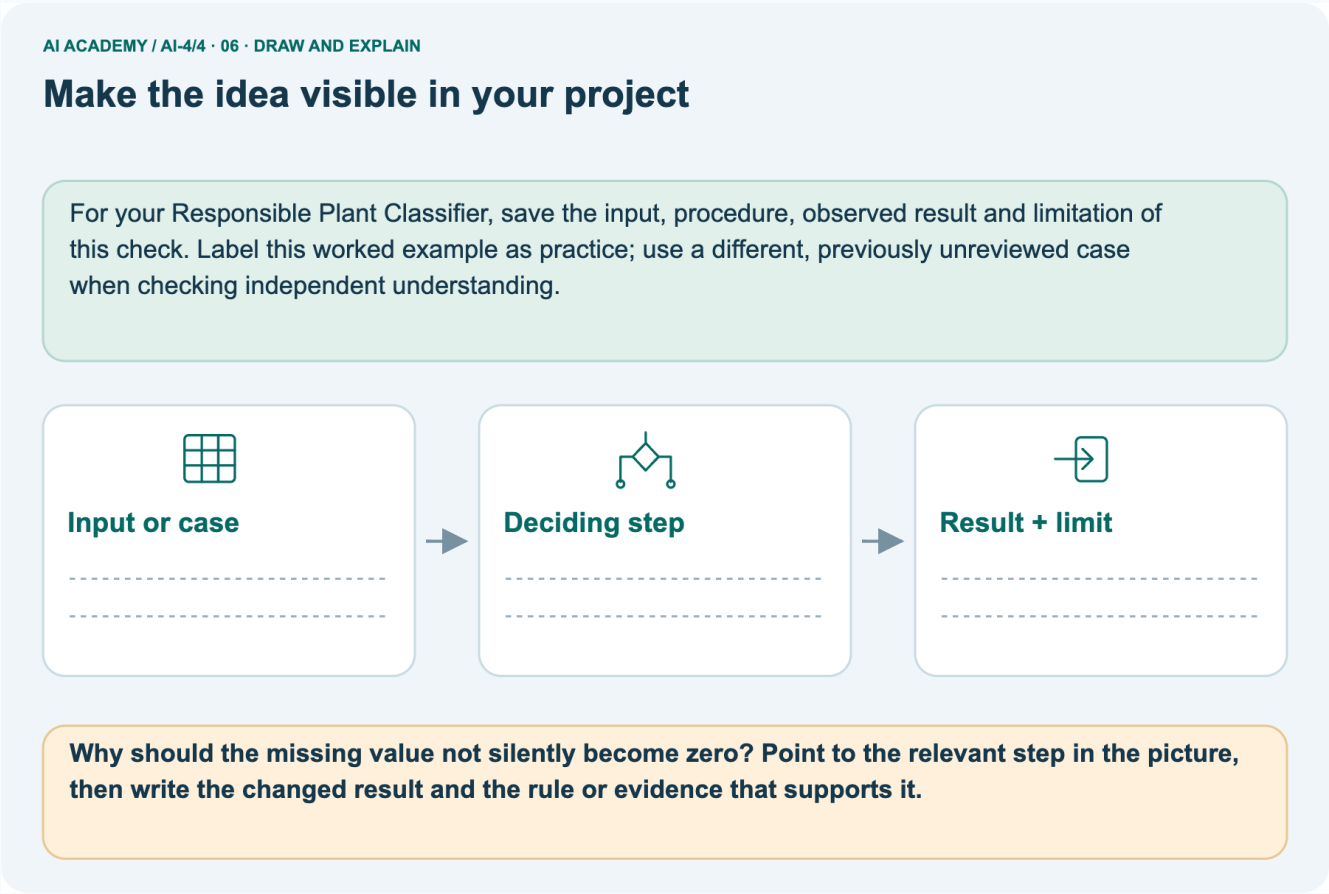
Read the picture in order. Report 3/4 measured, retaining the missing record separately. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: The measured-row mean may not represent the missing plants; investigate why those readings are absent.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should the missing value not silently become zero? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.